

ADARSH GAURAV

+91 7033375667 | adarshgaurav1624@gmail.com | [linkedin.com/in/adarsh-gaurav](https://www.linkedin.com/in/adarsh-gaurav) | github.com/adarshgaurav1951 | Bilaspur, India

PROFESSIONAL SUMMARY

Data Scientist and Computer Science undergraduate specializing in end-to-end Machine Learning pipelines, predictive modeling, statistical analysis, and Generative AI (RAG). Proficient in Python, SQL, PyTorch, and Scikit-learn, with proven success designing multimodal deep learning architectures achieving **96%+ classification accuracy** and sub-150ms REST API inference. Adept at high-dimensional feature engineering, Bayesian hyperparameter optimization (Optuna), and converting complex algorithmic outputs into data-driven business decisions.

TECHNICAL SKILLS

Machine Learning & Modeling:	Supervised & Unsupervised Learning, XGBoost, LightGBM, Random Forest, Logistic Regression, SVM, K-Means, Cross-Validation, Hyperparameter Tuning (Optuna, GridSearchCV), Model Evaluation (ROC-AUC, PR-AUC, F1)
Deep Learning & GenAI:	PyTorch, CNN, BiLSTM, Transformers, Multimodal Data Fusion, Retrieval-Augmented Generation (RAG), Vector Embeddings (BGE-M3), Semantic Search, OpenAI Whisper, Prompt Engineering, Ollama (Llama 3.2)
Data Analytics & Engineering:	Python, Pandas, NumPy, SQL (PostgreSQL, MySQL), Feature Engineering, Exploratory Data Analysis (EDA), Statistical Hypothesis Testing, Data Imputation, Matplotlib, Seaborn, Time-Series & Signal Processing
Developer Tools & Deployment:	FastAPI, Git, GitHub, Docker (Basics), Google Cloud Platform (GCP), RESTful APIs, Linux/Bash, VS Code, Google Colab

WORK EXPERIENCE

Data Science Intern | CodSoft

June 2025 – July 2025 | Remote

- Conducted end-to-end **Exploratory Data Analysis (EDA)** and feature engineering on 100K+ transactional records using **Python (Pandas, NumPy)**, uncovering key customer behavioral patterns and data distributions.
- Built and evaluated predictive classification models (Random Forest, XGBoost, Logistic Regression); resolved target imbalance via **SMOTE** and achieved an **88.4% F1-score** (+14.2% over baseline heuristics).
- Formulated 20+ predictive indicators and executed 5-fold cross-validation with grid search to optimize model convergence.
- Created interactive data visualizations with **Seaborn** and **Matplotlib** to communicate feature importances and analytical insights to non-technical stakeholders, driving data-informed business recommendations.

Software & Data Engineering Intern | CodeAlpha

Oct 2024 – Nov 2024 | Remote

- Engineered web applications and data capture interfaces with **JavaScript, HTML5, and Tailwind CSS**, optimizing responsive UI components and ensuring zero-loss client-side telemetry.
- Implemented rigorous client-side input validation and error handling schemas, reducing corrupted data submissions by **28%**.
- Optimized frontend asynchronous data fetching and API state management, reducing dashboard load times by **35%**.

SELECTED DATA SCIENCE & AI PROJECTS

Multimodal Deep Learning System for Clinical Disease Prediction

PyTorch, Scikit-learn, FastAPI, PostgreSQL

- Architected an end-to-end multimodal deep learning pipeline fusing continuous 12-lead ECG time-series signals with structured clinical laboratory records for early cardiovascular disease detection.
- Designed a hybrid **CNN-BiLSTM-Transformer** network with an attention-weighted feature bottleneck to jointly extract spatial waveform features and non-linear tabular interactions, beating unimodal tabular baselines by **15.6%**.
- Achieved **96.2% classification accuracy** and a **0.995 ROC-AUC** evaluated across 5-fold stratified cross-validation.
- Containerized the inference pipeline behind a high-throughput **FastAPI** REST service with PostgreSQL persistence, achieving sub-**150ms real-time** latency.

Autonomous RAG & Semantic Analytics Engine

Python, BGE-M3, Llama 3.2, Whisper, Scikit-learn

- Engineered an automated **Retrieval-Augmented Generation (RAG)** analytics engine to transcribe, index, and query multi-hour unstructured audio/video lectures and technical corpora without external cloud API dependencies.
- Automated speech-to-text pipeline using **OpenAI Whisper (large-v2)** and FFmpeg audio stream slicing, generating structured, time-stamped text chunks for semantic indexing.
- Built a dense vector retrieval index with 1024-dimensional **BGE-M3 embeddings** and cosine similarity search (Scikit-learn/NumPy), achieving sub-**45ms query response times**.
- Integrated local LLM inference via **Ollama (Llama 3.2)** with custom hallucination-mitigation prompt templates, ensuring 100% of responses include grounded, exact temporal video citations.

Customer Churn & Lifetime Value (LTV) Predictive Pipeline

Python, XGBoost, LightGBM, SQL, Optuna, SHAP

- Developed an end-to-end churn prediction pipeline on 250K+ customer profiles, engineering **30+ behavioral and Recency-Frequency-Monetary (RFM)** features using complex SQL window functions and Pandas.
- Trained and tuned **XGBoost and LightGBM** classifiers; mitigated severe class imbalance via **SMOTE-NC** and tuned hyperparameters with Bayesian optimization (**Optuna**), achieving **0.91 PR-AUC** and **86.5% Recall**.
- Leveraged **SHAP (SHapley Additive exPlanations)** for global and individual feature attribution, delivering interpretable churn risk factors to business stakeholders to proactively retain high-value accounts.

EDUCATION

Guru Ghasidas Vishwavidyalaya (A Central University) | Bilaspur, India

Dec 2022 – May 2026

Bachelor of Technology (B.Tech) in Computer Science and Engineering | **CGPA: 7.1 / 10**

Relevant Coursework: Machine Learning, Probability & Statistics, Data Structures & Algorithms, Database Management Systems (DBMS), Artificial Intelligence, Operating Systems, OOP in Java/C++

CERTIFICATIONS & LEADERSHIP

- Deloitte Data Analytics Job Simulation (Forage):** Completed corporate analytics engagement — executed data cleaning, exploratory data analysis (EDA), hypothesis testing, and designed executive data dashboards.
- Student Leader, Pregrad:** Directed campus tech initiatives and coordinated 15+ student teams across hackathons and software development sprints.